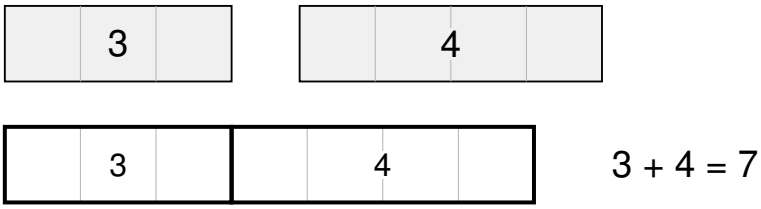
Put whole rods end to end. Leave no gaps or overlaps. You may use a length more than once. Changing the order of the same rods does not make a new way.



Problem 1: For 3-rods and 4-rods, find every impossible positive length. Explain why your list is complete.

1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	

Problem 2: Find the last impossible length for each pair: 3 and 5; 4 and 5. For each pair, give a finite collection of builds that settles every longer target.

3 and 5

4 and 5

Problem 3: Which lengths from 18 through 29 can be built with 4-rods and 7-rods? Convince someone without checking twelve separate cases.

18	19	20	21	22	23
24	25	26	27	28	29

Problem 4: Can 4-rods and 7-rods make 17? Explain your answer.

Problem 5: Which pairs have infinitely many impossible lengths: 2 and 4; 3 and 6; 4 and 6; 3 and 5; 4 and 7? Find a rule that separates them and explain it.

Problem 6: Find every way to build 24 with 3-rods and 4-rods. How can you change one way into another?

Problem 7: Predict the last impossible length for 5-rods and 7-rods, then test your prediction. Compare the answer with your earlier pairs and propose a rule that gives the last impossible length from the two rod lengths.

19	20	21	22	23	24	25
26	27	28	29	30	31	

Problem 8: Invent paper rods of two different whole-number lengths a and b that have no common divisor larger than 1. Explain why some finite group of builds must settle every length from some point on.